On the Perception and Adoption of Flakiness Scores

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I. Introduction

In 2018, Harman and O'Hearn called upon the scientific community to "Assume all Tests are Flaky", consequently establishing flaky tests as an inevitable risk for modern build pipelines[1].

Test flakiness scores are the first frontier to manage that risk. Gao et al. were pioneers in quantifying test flakiness, introducing the concept of $entropy$, with which they ascertained how external factors introduced variance to user-interface test run results [2]. Kowalczyk et al. proposed to model test flakiness as a function of it's results, defining $flip\ rate$, and utilising simple and weighted averages to aggregate flakiness scores [3], [4]. They were forerunners in evaluating their scoring model by comparing its results to the opinions of actual testers. Orthogonally, Rehman et al. defined test flakiness as the occurrence of test failures that do not lead to the identification of a product failure [5]. They introduced the No-Fault-Found rate and used it to map out the extent of functional test flakiness for Radio Base Station (RBS) software, ultimately affecting the testing practices of the test teams within Ericsson with whom they worked.

II. Novel Perspective

Earlier work has focussed on the two primary outcomes, Pass and Fail, and the application of test flakiness scores for unit and integration tests. Nevertheless, a greater variety of test outcomes exists, as has been noted in the literature by Malmir et al. and Wendler et al. [6], [7], as well as levels of testing [8].

Within our study, we consider system tests because our industrial partners asserted they do not tolerate flakiness at the unit level. We have observed, from their datasets, that they too consider more test outcomes, (Pass, Fail, Error, Skip). To facilitate these test outcomes, we extended Kowalczyk et al.'s definition of $flip\ rate$ to $transition\ rate$, [9], which considers the observed rate of each individual transition for a given number of test runs per version of the system under test. This definition allows for any number of test outcomes. Derived from this, $multi\text{-}transition\ rate$ is the total flakiness of the test for that version. In our paper [9], we demonstrate using two artificial datasets, how this score can support 3 strategies for addressing flakiness; Fix, Rerun and Monitor.

Test flakiness has been shown to affect the entire build pipeline [10], which translates to the DevOps roles along that pipeline [11]. However, to the best of our knowledge, a role-based investigation into practitioner perception of test flakiness scores is yet to be conducted. This study aims to open up the discussion about the usability of test flakiness scores across roles further bridging the gap between research and practice, providing valuable insights for tool creators and uncovering future research directions.

III. Work In progress

As a follow-up to our simulation with artificial datasets, we aim to shed more light on the practical

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implications of flakiness scores from an industrial context. We have confirmation of two industrial partners who intend to participate in a pilot study concerning the extended flakiness score proposed in [9]. We convinced our partners to participate in this study through a series of discussions, preliminary result presentations and tool demonstrations based on open source software.

This study is driven by two principal questions.

- **RQ1:** *Perception.* How do practitioners perceive the current flakiness of their test suites?
  - **Motivation.** We must ensure that we capture the similarity of flakiness scores to test flakiness perception per role along the DevOps pipeline.
  - **Approach.** We will start by ascertaining the teams experience so far with test flakiness scores. We will generate the scores of their datasets using our scoring tool. We will ask the industrial teams to rate the flakiness of their tests based on their experience. For this, we will provide a 5 point Lickert scale from Not Flaky to Extremely Flaky. We will then compute the correlation of their perceptions with the scores provided by the scoring tool per role.
- **RQ2:** *Adoption.* How do the teams adopt the flakiness scores to their own situations?
  - **Motivation.** We must gain a broader understanding of how industrial teams are applying flakiness scores, especially at the level of system testing.
  - **Approach.** We will establish the teams experience adopting a test flakiness score particularly focussing on testing practices, environment, culture and the roadblocks they have encountered.

Both research questions will be answered within role-based focus groups. For a more general understanding, we shall compare the answers of the different focus groups and compute their similarity using Natural Language Processing techniques.

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